

AYUSH AGGARWAL

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SUMMARY

Software Engineer with **4+ years of experience** building **backend services, cloud infrastructure, and data platforms on AWS**. Experienced with **Node.js, Python, React, and serverless architectures**, including analytics systems built with **Athena and Metabase**. Interested in **backend, platform, and cloud infrastructure engineering**.

WORK EXPERIENCE

Ellucian

Software Engineer

Reston, VA

Jun 2024 - Present

- Designed the end-to-end architecture and **access-control model** for a **lakehouse analytics platform**, migrating reporting workloads from a **Postgres-based data lake** to **S3 Iceberg tables** queried through **Athena**, with governance enforced through **IAM** and **Lake Formation**.
- Defined **secure request flows** and **service-to-service authentication** patterns between internal services, **Metabase**, **Athena**, and AWS resources, collaborating with **AWS SMEs** on **Lake Formation**, **S3 Tables**, and permission design.
- Analyzed complex **SQL reporting dependencies** and migration paths across replicate tables, composite tables, reporting views, and **BI dashboards** to support the transition from Postgres to the lakehouse architecture.
- Designed and deployed **full-stack product features** using **React**, **Python APIs**, and **AWS serverless services**, supporting customer-facing and internal analytics workflows.
- Led the **Node.js 22 upgrade** across multiple services, resolving **CI compatibility issues** and enabling adoption of the newer runtime across engineering teams.
- Built a **Lambda-based automation pipeline** to triage **Datadog logs**, reducing manual operational effort by **50%** and accelerating incident investigation.
- Developed **Playwright-based UI/API automation** and **self-service Jenkins integrations**, improving release confidence, regression coverage, and developer productivity.

Software Engineer Intern

May 2023 - Aug 2023

- Built a **self-service Insights Card platform** using **React**, **AWS Lambda**, and **Terraform-managed APIs**, enabling **57 colleges** to create analytics dashboards without engineering support.
- Implemented **WebSocket APIs** and **SQS-based messaging** backed by **DynamoDB streams** to deliver **real-time pricing and availability updates**.

ZS Associates

Software Engineer

Gurgaon, India

Aug 2020 - Jun 2022

- Migrated legacy **MySQL/Oracle** pipelines to AWS (**S3/Athena/Redshift**), improving query performance **~30%** and integration throughput **~20%**; introduced partitioning + Parquet for cost/perf.
- Built data apps and APIs that pushed modeled data into downstream tools; added job orchestration, retries, and monitoring for reliable **syncs**.
- Partnered with business stakeholders to define segments/filters and publish metrics used for experiments and campaigns.

Info Space Export

Software Developer Intern

Gurgaon, India

May 2019 - Jul 2019

- Developed features for an **online gaming platform supporting 35k+ users** using **PHP, JavaScript, and Oracle SQL**.

NIIT Ltd

Software Developer Intern

Gurgaon, India

May 2018 - Jul 2018

- Designed student performance dashboards for internal use, utilizing **D3.js** and JavaScript data charting libraries.
- Developed dynamic watermarking software for PDFs using **C#**, enhancing security protocols by **55%** and reinforcing backend data protection measures.

Define Labs

Software Developer Intern

Gurgaon, India

May 2017 - Jul 2017

- Built a scalable in-house user management web app, showcasing full-stack development expertise and adherence to technical best practices.

SKILLS

Languages: Python, C/C++, Java, JavaScript, Bash scripting, SQL

Backend and Data: Node.js, REST APIs, PySpark, Data Pipelines

Frontend: React.js, Typescript, HTML, CSS

Cloud: AWS (Lambda, S3, Athena, IAM, DynamoDB, SQS, SNS), Terraform, Docker

Tools: Git, Jenkins, Datadog, Playwright

EDUCATION

Virginia Tech, Blacksburg, VA, USA

Aug 2022 - May 2024

Master of Engineering, Computer Science

GPA: 3.96/4

- **Achievements:** Graduate Research Assistant: AI in Design, UI/UX in Figma, Full Stack Development
- **Coursework:** Web Application Dev, Cloud Computing, Software Eng., Mobile App Development, Large Scale Systems Data Analysis

GGSIIP University, Delhi, India

Aug 2016 - Jul 2020

Bachelor of Technology, Information Technology

GPA: 9.12/10

- **Achievements:** Secured first rank in University Examination 2017.
- Served as a mentor for Data Structures and Algorithms for two years.

PROJECTS

- Engineered an AI-Powered Web-Based Coding Platform (ReactJS, NodeJS, OpenAI) featuring real-time code optimization and intelligent suggestions to aid students in interview preparation.
- Utilized NLP and LLMs to generate automated lectures, reducing course preparation time by 40% and boosting educator efficiency.

LEADERSHIP & RECOGNITION

- Graduate Research Assistant: AI in Design, UI/UX in Figma, Full Stack Development
- Secured first rank in University Examination 2017. Served as a mentor for Data Structures and Algorithms for two years.
- **Awards:** "Impact and Recognition" Award - for outstanding client impact, "Celebrating Impact - Best Team" Award